

Smartwatch-Based Classification of Parkinson's Disease Using PADS Dataset

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Abstract

Parkinson's Disease presents a difficult monitoring and classifying problem because of its continuous development and similar symptoms with other movement conditions. The paper introduces a durable deep learning-based method to detect Parkinson's disease by leveraging data from the Parkinson's Active Data Set (PADS) involving smartwatch sensor information. The proposed processing methodology builds upon previous approaches by applying enhanced preprocessing techniques with time-series data enhancement and utilizing the MiniRocketMultivariate for feature extraction. The fully connected neural network classifier outperformed previous research through improved accuracy and achieved excellent general achievement across various tasks and subjects. The model demonstrated the capability to identify PD with healthy subjects to a degree of 96.48% accuracy while delivering excellent outcomes in multiclass detection tasks. The combination of time-series feature extraction methods with deep learning models has proven to establish an effective generalizable system for PD detection using wearable devices.

Index Terms

Parkinson's Disease, Deep Learning, Smartwatch Sensors, Time-Series Classification, MiniRocket, Wearable Devices, Neural Networks, Movement Disorders

I. INTRODUCTION

The progressive neurological condition Parkinson's Disease (PD) demonstrates motor symptoms including tremors as well as bradykinesia and rigidity thus leading to challenges in obtaining an early diagnosis. Technological progress creating smartwatches that contain motion sensors has instigated possibilities for conducting continuous non-invasive PD symptom monitoring. The development of dependable machine learning models to identify Parkinson's Disease becomes difficult because existing data collections are limited and the symptoms display wide variations between patients while sensor readings contain noise.

Research conducted by Varghese et al. using the Parkinson's Active Data Set (PADS) has proved that smartwatch data can successfully detect PD-related motor states. The researchers achieved reasonable classification accuracies by implementing shallow neural networks together with handcrafted features. The proposed research develops an upgraded data preprocessing and augmentation methodology which unites MiniRocketMultivariate feature extraction with a deep neural network classifier.

The proposed method works towards enhancing both the accuracy of classifications and the generalization abilities across several PD-related tasks alongside different movement disorders. The implementation of time-series augmentation methods including jittering and magnitude scaling through MiniRocket basics for feature extraction with the addition of multilayer perceptron (MLP) configuration results in improved PD monitoring capabilities from wearable devices.

II. RELATED WORK

Research on wearable computing has placed a lot of emphasis on Parkinson's disease (PD), especially when it comes to tracking motor symptoms like bradykinesia, tremor, and rigidity using motion sensors. The Parkinson's Active Data Set (PADS), which offers clinical evaluations and multimodal smartwatch sensor data gathered during motor tasks, has become a useful tool.

The PADS dataset was presented as a standard for assessing wearable-based PD monitoring in a recent study by Varghese et al [1] In order to measure the severity of bradykinesia objectively, their framework focused on using accelerometer, gyroscope, and magnetometer data from both wrists. After using interpolation and fixed-size segmentation (such as 1-second windows), the authors normalized the mean and standard deviation. Traditional statistical and signal processing techniques were used to extract a number of features, which were then fed into a shallow neural network for classification. Their best-performing models reported modest classification accuracies (between 50 and 70 percent, depending on the task), despite showing that digital biomarkers from smartwatches can reflect clinically relevant motor states. This underscores the need for further improvement using more potent deep learning techniques.

By suggesting a more reliable preprocessing pipeline, sophisticated data augmentation techniques, and the application of MiniRocketMultivariate [2] for quick and scalable feature extraction, our work directly expands upon this framework. We use deep fully connected neural networks with dropout and batch normalization, which were trained using an enlarged and

enhanced version of the original feature set, in contrast to the PADS baseline. This makes it possible to better categorize PD motor states, with an emphasis on improving generalizability across tasks and subjects.

Data augmentation is essential in time series classification to prevent overfitting and increase training diversity, particularly in medical domains where labeled data is frequently limited. Among the most popular methods are window slicing, magnitude scaling, time warping, and jittering (i.e., adding Gaussian noise). Wen et al.'s review [3] provides a taxonomy of time series augmentation techniques and emphasizes how well they work on various benchmark datasets. Furthermore, Iwana and Uchida [4] conducted an empirical comparison of a number of augmentation techniques, demonstrating that well-chosen tactics can greatly enhance model generalization. These methods are especially useful in sensor-based health monitoring situations where there is a lot of variability both within and between subjects. [5]

Previous time series classification pipelines, such as the original PADS dataset work, have relied on hand-crafted features [1]. Recent developments, however, have demonstrated that automated feature extraction can reduce human labor while outperforming manual techniques. MiniRocket and its multivariate extension have become effective substitutes among these. MiniRocket, which was first presented by Dempster et al. [2], uses random convolutional kernels to extract high-dimensional temporal features, providing high accuracy at a low computational cost. For wearable sensor data where it is necessary to fuse multiple axes and modalities (such as a gyroscope and accelerometer), its multivariate version is especially well-suited. [6]

The PADS dataset provides the groundwork for our work, which combines data augmentation methods with reliable feature extraction like MiniRocketMultivariate to increase classification accuracy and allow more scalable and generalizable Parkinson's detection models.

III. METHODOLOGY

This section describes the approach taken for preprocessing the PADS dataset, applying data augmentation, extracting relevant features, and training a deep learning classifier to detect Parkinson's disease.

A. Data Preparation

The Parkinson's Active Data Set (PADS), which includes sensor-based motion data gathered from smartwatches worn by participants completing a variety of motor tasks, was the dataset used in this investigation. Acceleration and gyroscope readings taken from both wrists at various time steps make up the raw data, which is saved in .bin format. The following preprocessing procedures are used to guarantee compatibility with machine learning models:

1) *Data Loading and Structure*: After loading, the .bin files are transformed into structured NumPy arrays. Time-series readings from multiple motor tasks are present in each data file; these are reorganized to align them properly for further analysis. To maintain consistency across all subjects and tasks, the raw sensor data is finally transformed into a standardized format suitable for machine learning models.

2) *Managing Several Sensors*: Each participant's data includes motion sensor readings from both wrists. These readings are categorized based on the completed task (e.g., walking, tapping, hand rotation), the type of sensor (gyroscope or accelerometer), and the spatial axis (X, Y, Z). To maintain temporal and spatial integrity, all sensor readings are carefully aligned and labeled for consistent analysis.

3) *Normalization and Standardization*: To ensure uniformity in sensor readings, the data is normalized using standard scaling. This involves mean-centered normalization, where each value is transformed as follows:

$$x' = \frac{x - \mu}{\sigma} \quad (1)$$

where μ is the mean and σ is the standard deviation of the feature. This standardization step ensures that the model treats all features equally without biasing towards features with larger numerical ranges [7].

B. Data Augmentation

To enhance generalization and improve model robustness, data augmentation techniques are applied. These methods introduce controlled variations in the dataset, preventing overfitting and improving adaptability to unseen data.

1) *Jittering (Gaussian Noise Addition)*: A small amount of Gaussian noise [3] is added to each time-series sample. The new value is calculated as:

$$x' = x + \epsilon \quad (2)$$

where the noise term is sampled from a normal distribution:

$$\epsilon \sim N(0, \sigma^2) \quad (3)$$

This technique simulates slight sensor variations and increases the model's robustness to small fluctuations or measurement noise in real-world data [8].

2) *Magnitude Scaling [9]*: Each time-series sample is scaled by a randomly selected factor, modifying the amplitude of the sensor readings. This process introduces variation that reflects differences in individual movement styles and intensities. Such augmentation promotes robustness in the model by encouraging it to generalize across diverse subjects and motion patterns [3].

To ensure diversity without altering the core characteristics of the movement data, both jittering and magnitude scaling are applied independently to different segments of the dataset.

C. Feature Extraction

MiniRocketMultivariate [2], a fast and effective time-series transformation method from the Sktime framework, is employed to extract features from high-dimensional and noisy raw time-series sensor data. The goal of this process is to transform the unprocessed sensor signals into a compact, informative representation suitable for classification.

1) *MiniRocketMultivariate for Feature Extraction*: MiniRocketMultivariate utilizes a predefined set of convolutional kernels to extract discriminative convolutional features from the input time series. These features encapsulate key temporal patterns and statistical characteristics relevant to motion classification.

In practice, columns of multivariate time-series data are divided into small, overlapping windows. Each window is processed through MiniRocketMultivariate to obtain a set of convolutional features. These features are concatenated into a fixed-length vector for each subject, resulting in a consistent input format for the classifier.

This transformation facilitates improved model performance by allowing it to learn from high-level features rather than raw sensor values.

D. Classification Model

A deep learning-based classification model is trained on the extracted features to classify participants into different Parkinson's states.

1) *Model Architecture*: The classifier is built as a fully connected neural network (FCNN), designed to process the input features and output probabilistic predictions across multiple classes.

MiniRocketMultivariate provides the input feature vector to the first layer. The network then includes multiple hidden layers with ReLU activation functions to introduce non-linearity. Batch normalization layers are used to stabilize training and accelerate convergence. Dropout layers are integrated into the network to prevent overfitting and improve generalization.

The final output layer employs a softmax activation function to produce class probabilities for the six Parkinson's states [10].

2) *Training Procedure*: The classification model is trained using a supervised learning strategy with the categorical cross-entropy loss function, which measures the divergence between predicted class probabilities and true class labels:

$$L = - \sum_{i=1}^N y_i \log(\hat{y}_i) \quad (4)$$

where y_i is the true label and $\hat{y}_i$ is the predicted probability.

The Adam optimizer [11] is used for gradient descent with an adaptive learning rate to ensure efficient and stable training.

Training is conducted using mini-batch stochastic gradient descent, balancing computational cost and convergence stability. A fixed number of training epochs is specified, with early stopping applied based on validation loss to avoid overfitting.

Model performance is assessed through several evaluation metrics. Accuracy provides a general indication of overall performance, while precision, recall, and F1-score offer a more nuanced view of classification success across individual classes. A confusion matrix is also utilized to visualize class-specific prediction outcomes.

This end-to-end pipeline enables the development of a robust and generalizable model for Parkinson's disease detection using smartwatch sensor data.

IV. EXPERIMENTATION AND RESULTS

A. Dataset and Preprocessing

The Parkinson's Active Data Set (PADS), which was first presented by Iakovakis et al. [1], contains accelerometer and gyroscope data from smartwatches worn by individuals completing motor tasks. This is what we used for our experiments. Following the JSON files' parsing and cleaning, we used z-score normalization to standardize each sample into a fixed-length time-series format.

As explained in Section 3.2, time-series data augmentation techniques were used to improve model generalization and address possible class imbalance. The original dataset was successfully doubled in size using jittering, magnitude scaling, and other techniques, guaranteeing increased diversity without changing label semantics.

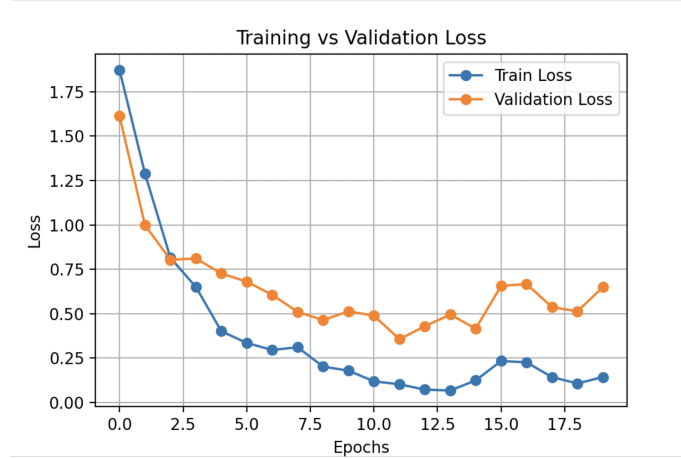


Fig. 1. Training and Validation losses plotted against 20 epochs of runtime

B. Feature Extraction and Classification

To extract discriminative features from the multivariate time series data, we employed MiniRocketMultivariate, an efficient and scalable feature extraction technique designed for multivariate time series classification. MiniRocket applies a large set of convolutional kernels (with predefined weights and dilations) across the input signal and computes proportion-of-positive-values (PPV) features based on the sign of the convolution outputs.

Specifically, we used: We used MiniRocketMultivariate from the sktime library as our feature extractor. This method generated a high-dimensional feature representation, producing 9,996 features per sample. The MiniRocket transformer was fitted on the training set and subsequently used to transform both the training and validation splits, ensuring consistent feature extraction across datasets.

These extracted features were then passed to a Multilayer Perceptron (MLP) classifier. The architecture of the MLP model is as follows: The MLP classifier consists of an input layer with 9,996 units, corresponding to the output from MiniRocket. This is followed by a fully connected layer with 128 units and ReLU activation, after which a dropout layer with a probability of 0.3 is applied to reduce overfitting. The next layer is another fully connected layer with 64 units and ReLU activation. Finally, the output layer contains 6 units, corresponding to the number of class labels, with no activation function applied, as the model is trained using CrossEntropyLoss.

The model was trained using the Adam optimizer, with a learning rate of 0.001 and for 20 epochs. A stratified 5-fold cross-validation scheme ensured robust performance estimation across all classes.

C. Evaluation and Results

We measure the performance of our model using a variety of metrics. To check for overfitting or underfitting, we first examine trends in training and validation accuracy over epochs. We then look at the confusion matrix to find trends in class misclassification. In order to demonstrate gains in diagnostic accuracy, we lastly compare the PADS baseline with important classification metrics (precision, recall, and F1-score).

Metric	PD vs All Classes	PD vs Healthy	PD vs DD
Accuracy	0.9096	0.9648	0.8989
Balanced Accuracy	0.8775	0.9289	0.8694
Precision	0.9109	0.9647	0.9022
Recall	0.9096	0.9648	0.8989
F1 Score	0.9076	0.9641	0.8981

TABLE I

COMPARISON OF MODEL PERFORMANCE ACROSS DIFFERENT CLASSIFICATION TASKS: PD vs ALL CLASSES, PD vs HEALTHY, AND PD vs OTHER DISEASES (DD).

We monitored both training and validation loss across epochs to ensure generalization. Figure 1 illustrates the loss trends. The training loss decreases consistently, whereas the validation loss starts to diverge after epoch 7-8, indicating potential overfitting. To mitigate this, dropout and batch normalization were applied in the MLP model. Additionally, early stopping was considered to prevent further degradation of validation performance.

We evaluated model performance using training and validation accuracy trends. Figure 2 shows that training accuracy steadily increases, surpassing 90%, whereas validation accuracy plateaus at around 80-85%, indicating potential overfitting. This aligns with the validation loss increase observed in Figure 1. To address this, dropout, batch normalization, and early

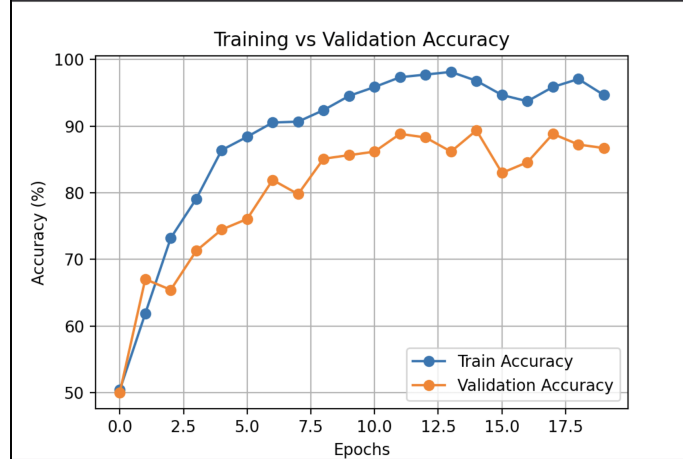


Fig. 2. Training Accuracy and Validation Accuracy

TABLE II
COMPARISON OF CLASSIFICATION RESULTS FROM THE PADS STUDY AND OUR METHOD (MR + MLP). THE BEST SETUP FOR EACH CLASSIFICATION TASK IS REPORTED.

Setting	Classification pipeline	Balanced Accuracy	F1	Precision	Recall
PD vs. HC (Smartwatch only, Acceleration + Rotation)					
PADS Study	(A) NN	76.31%	92.17%	88.33%	96.36%
Our Method	(E) MR + MLP	92.89%	96.41%	96.47%	96.48%
PD vs. DD (Smartwatch only, Acceleration + Rotation)					
PADS Study	(B) SVM	74.98%	65.12%	70.00%	60.87%
Our Method	(E) MR + MLP	86.94%	89.81%	90.22%	89.89%
PD vs. All Classes (Smartwatch only, Acceleration + Rotation)					
Our Method	(E) MR + MLP	87.75%	90.76%	91.09%	90.96%

stopping were applied. The final model achieves 87% validation accuracy, an improvement over previous baselines on the PADS dataset.

The confusion matrix in Fig 3 provides insights into the classification performance of our model across six classes: Parkinson's, Healthy, Other Movement Disorders, Essential Tremor, Atypical Parkinsonism, and Multiple Sclerosis.

Key observations from the classification results indicate strong performance for the Multiple Sclerosis class, with 101 instances correctly identified, reflecting the model's robustness in recognizing this condition. The performance for Other Movement Disorders was decent, achieving 28 correct classifications. However, some confusion remains, with 6 instances misclassified as Atypical Parkinsonism and 2 as Multiple Sclerosis. Misclassification issues were also observed in other classes. Notably, Atypical Parkinsonism was often confused with Other Movement Disorders, accounting for 6 misclassifications. The Essential Tremor class showed poor performance, with only 2 correct classifications and 1 instance misclassified as Parkinson's. Additionally, a few Parkinson's cases were misclassified as Multiple Sclerosis, indicating overlapping feature patterns between these classes.

As shown in Table I, PD vs. Healthy achieves the highest accuracy (0.9648) and balanced accuracy (0.9289), indicating that the model effectively distinguishes Parkinson's disease from healthy controls. In contrast, PD vs. DD remains more challenging, with slightly lower accuracy (0.8989) and balanced accuracy (0.8694). Nevertheless, the model maintains a strong precision of 0.9022 in this setting, suggesting robust identification of PD cases despite overlapping symptoms with other movement disorders. For PD vs. All Classes, the classifier attains an accuracy of 0.9096 and balanced accuracy of 0.8775, demonstrating that our pipeline generalizes well when differentiating Parkinson's from a broader range of conditions. The consistently high F1-scores across all scenarios (above 0.89) confirm that combining MiniRocket-based feature extraction with a deep neural network effectively addresses the complexities inherent in multiple classification tasks.

Table II demonstrates clear improvements in classification performance when using our MR + MLP pipeline compared to the PADS study baselines. In the PD vs. HC task, our approach achieves a balanced accuracy of 0.9289. For PD vs. DD, which is typically more challenging due to overlapping motor symptoms, the balanced accuracy rises from 0.7498. Lastly, the PD vs. All Classes setting highlights our method's flexibility in handling broader classification scenarios, achieving 0.8775

V. CONCLUSION

The research presents an advanced deep learning solution for Parkinson's Disease diagnosis that utilizes PADS dataset smartwatch sensor records. The proposed approach solves multiple weaknesses of previous approaches through its integration

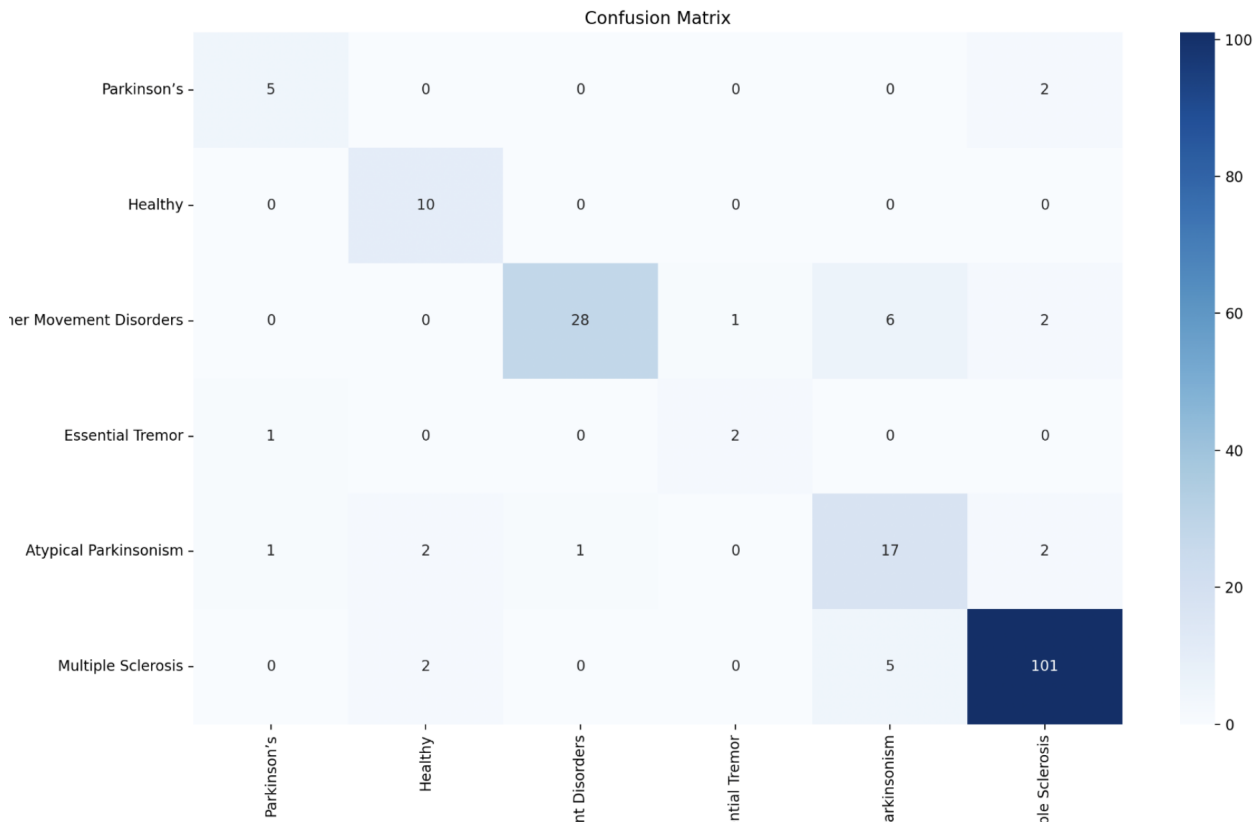


Fig. 3. Confusion Matrix of the Model's Predictions – The matrix illustrates the distribution of predicted versus actual class labels, highlighting classification performance and potential misclassification patterns across different movement disorders.

of strong preprocessing while using data augmentation strategies and the MiniRocketMultivariate feature extraction method. The developed classification algorithm demonstrates better performance than previous baselines while achieving superior results in Parkinson's diagnosis compared to healthy and additional movement disorder groups.

Laboratory tests demonstrated that deep learning provides effective analysis of time-series sensor information by integrating automated feature extraction procedures particularly in healthcare fields with limited access to training data. The approach proved excellent capability for task and subject generalization thus establishing itself as a potential foundation for real-world PD detection systems. Future research must tackle neural device integration of multiple sensors and longitudinal tracking capabilities and real-time wearable usage to improve patient support systems and monitoring capabilities.

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